

João Roque Costa

MSC GRADUATE, COMPUTER SCIENCE AND ENGINEERING

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A Master's graduate with a strong interest in **problem-solving**, **financial markets**, **artificial intelligence**, and **data science**; eager to apply my theoretical knowledge in practical settings. Looking to join a dynamic team, develop new skills, and tackle real challenges in the IT industry.

Professional Experience

Instituto Superior Técnico

Lisbon, Portugal

Invited Teaching Assistant

Oct. 2024 - Jul. 2025

- Taught and evaluated practical classes on Databases and Data Science.
- Assisted students with hands-on projects and provided feedback on their progress.
- Lectured on database design and management using SQL within PostgreSQL and pgAdmin.
- Instructed students in data treatment, visualization, and analysis techniques using Python (pandas/matplotlib).

Education

Instituto Superior Técnico

Lisbon, Portugal

M.Sc. Computer Science and Engineering

Sep 2023 - Nov 2025

- Specialized in **Data Science** with emphasis in **Artificial Intelligence**, maintaining a focus on scalable data architectures and distributed systems.
- Relevant Coursework: Information Visualization, Data and Information Systems Management, Data Analysis and Integration, Machine Learning, Deep Learning, Cloud Computing and Virtualization, Optimization Algorithms.
- Thesis: Maximum Variance Unfolding on Disjoint Manifolds (MVU-DM).
 - Developed a novel extension to the MVU algorithm that enables nonlinear dimensionality reduction on disconnected datasets.
 - Engineered a decomposition-based approach that allows for parallel processing, significantly improving scalability for large-scale datasets that were previously computationally infeasible.
 - Achieved equal-quality embeddings and preservation of local geometric structures compared to the standard method.

Instituto Superior Técnico

Lisbon, Portugal

B.Sc. Computer Science and Engineering

Sep 2020 - Jul 2023

- Developed a strong foundation in **Distributed Systems**, **Operating Systems**, and **Database Theory**.
- Collaborated on several full-stack and systems-level projects, emphasizing memory management in C/C++ and relational schema design in SQL.

Extracurricular

Personal Software Projects (Self-Hosted)

- Engineered an automated betting bot for parimutuel systems using a modified Kelly Criterion for risk management and optimal stake sizing; integrated Faceit and Telegram APIs for real-time data and alerts.
- Developed a custom Discord-based matchmaking and ELO ranking system, utilizing automated queue management.
- Architected a multi-platform arbitrage engine and inventory manager, implementing automated price monitoring and trade execution across disparate marketplaces.
- Built a full-stack data visualization dashboard to analyze player performance metrics, converting raw API data into actionable strategic insights.

Skills & Competencies

Programming	Python (Scikit-learn, TensorFlow, Keras, Pandas, NumPy, Matplotlib), Java, C++, C#
Databases	SQL, Microsoft SQL Server (SSMS), PostgreSQL, MySQL, pgAdmin
Data Engineering	Pentaho PDI, SQL Profiler, Tuning Advisor (DTA), DataCleaner tool
Business Intelligence	Pentaho (Server, Schema Workbench, Report Designer), Saiku Analytics
Cloud & Architecture	AWS, ArchiMate, Camunda Modeler, Enterprise Architecture (Sparx)
Developer Tools	Git, Linux, Bash Scripting, RESTful APIs, LaTeX
Languages	Portuguese (Native), English (Fluent)